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Insert-molded filter and production method for insert-molded filter

Abstract

This insert-molded filter is obtained by insert-molding a cylindrically-shaped mesh into a resin body. This insert-molded filter is provided with: a body which is equipped with a first annular part, a second annular part, and multiple columnar parts connected to the first and second annular parts, and which has multiple openings formed so as to be separated by the multiple columnar parts between the first and second annular parts; and a mesh which is integrally molded with the body so as to cover the multiple openings separated by the multiple columnar parts. The body is provided with a resin introduction part that is disposed outside of the mesh in the radial direction of the first annular part.

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Background/Summary

RELATED APPLICATIONS

(1) The present invention is a U.S. National Stage under 35 USC 371 patent application, claiming

priority to Serial No. PCT/JP2021/005018, filed on Feb. 10, 2021, which claims priority from Japanese Patent Application No. 2020-029339 filed on Feb. 25, 2020; the entireties of both are hereby incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to an insert-molded filter and a method for manufacturing an insert-molded filter.

BACKGROUND ART

(3) Insert molding is used to manufacture a filter for fluid. In one example of insert molding, a mold includes a lower mold that has a cylindrical core pin and an upper mold that has a hole into which the core pin is inserted. Ribs are arranged on the bottom of the hole in the upper mold at equal intervals in the circumferential direction of the core pin. The ribs are inclined relative to the central axis of the core pin.

(4) When performing insert molding with such a mold, a tubular mesh is prepared. The tubular mesh is arranged on the outer surface of the core pin. In this state, the core pin, together with the mesh, is inserted into the hole of the upper mold. The parts of the distal end of the mesh contacting the ribs are moved further toward the central axis of the mesh than the parts of the mesh that do not contact the ribs. As a result, the distal end of the mesh is partially embedded in a body formed by resin injected into the mold. This increases the separation resistance of the mesh from the body compared to when the distal end of the mesh is in contact with the outer surface of the body (refer to, for example, Patent Document 1).

CITATION LIST

Patent Literature

(5) Patent Literature 1: Japanese Laid-Open Patent Publication No. 2018-69114

SUMMARY OF INVENTION

Technical Problem

(6) In the above insert molding, parts of the distal end of the mesh need to contact the ribs to increase the separation resistance of the mesh from the body. The positions of the ribs impose restrictions on the structure of the mold. This, in turn, imposes restrictions on the position of the mesh in the completed filter. Thus, the degree of freedom for the position of the mesh on the filter needs to be increased while ensuring the separation resistance of the mesh from the body.

(7) It is an objective of the present disclosure to provide an insert-molded filter that ensures the separation resistance of a mesh from a body and increases the degree of freedom for the position of the mesh in a filter, and a method for manufacturing the insert-molded filter.

Solution to Problem

(8) According to one aspect of the present disclosure, an insert-molded filter includes a plastic body including a first ring, a second ring, and posts that are connected to the first ring and the second ring, where the body includes openings partitioned by the posts between the first ring and the second ring; and a mesh including a tubular portion, the mesh being molded integrally with the body through insert molding to cover the openings partitioned by the posts. The body includes a plastic inlet portion positioned outward from the mesh in a radial direction of the first ring.

(9) According to another aspect of the present disclosure, a method for manufacturing an insert-molded filter includes preparing an insert-molded mesh including a tubular portion; arranging the insert-molded mesh in a mold; and forcing resin through a gate, located outward from the tubular portion in a radial direction of the tubular portion, into a molding portion of the mold where a body is molded integrally with the insert-molded mesh.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a perspective view of an insert-molded filter according to one embodiment.
- (2) FIG. 2 is a plan view of the insert-molded filter shown in FIG. 1 as viewed from a view point that is opposed to a first ring.
- (3) FIG. 3 is a side view shown in the insert-molded filter of FIG. 1 as viewed from a view point that is opposed to one post.
- (4) FIG. 4 is a perspective view shown in an insert-molded mesh used to manufacture the insert-molded filter of FIG. 1.
- (5) FIG. 5 is a diagram illustrating the operation of an insert-molded filter.
- (6) FIG. 6 is a cross-sectional view showing the structure of the insert-molded filter of FIG. 1.

DESCRIPTION OF EMBODIMENTS

- (7) An insert-molded filter **10** according to one embodiment will now be described with reference to FIGS. 1 to 6.
- (8) As shown in FIG. 1, the insert-molded filter **10** includes a body **11** and a mesh **12**. The mesh **12** is insert-molded to the plastic body **11** in the insert-molded filter **10** and includes a bottom wall (end wall) **12B** and a tubular portion **12A** connected to a portion of the bottom wall **12B**.
- (9) The body **11** includes a first ring **11A**, a second ring **11B**, and posts **11C**. The posts **11C** are connected to the first ring **11A** and the second ring **11B**. The body **11** includes openings **11E1** partitioned by the posts **11C** between the first ring **11A** and the second ring **11B**. The mesh **12** is molded integrally with the body **11** so as to cover an opening **11E2** in the first ring **11A** and the openings **11E1** partitioned by the posts **11C**. The body **11** includes plastic inlet portions **11D** that are positioned outward from the mesh **12** in the radial direction of the first ring **11A**.
- (10) When manufacturing the insert-molded filter **10**, molten resin is drawn through gates, which are located in correspondence with where the inlet portions **11D** of the completed insert-molded filter **10** will be located, into portions of the mold where the body **11** is molded. The gates are positioned outward from the mesh **12** in the radial direction of the first ring **11A**. Thus, the resin forced through the gates bends the tube end of the mesh **12** inward in the radial direction of the first ring **11A**. Consequently, in the completed insert-molded filter **10**, at least part of the tube end of the mesh **12** is embedded in the first ring **11A**. This ensures the separation resistance of the mesh **12** from the body **11**. The positions of the gates do not impose restrictions on the position of the mesh **12** in the insert-molded filter **10**. This increases the degree of freedom for the position of the mesh **12** in the insert-molded filter **10**.
- (11) The inlet portions **11D** are located on the posts **11C**. This allows the inlet portions **11D**, which are located on the posts **11C**, to be easily positioned outward from the mesh **12** in the radial direction of the first ring **11A**.
- (12) The first ring **11A** and the second ring **11B** are each annular. The diameter of the second ring **11B** is greater than the diameter of the first ring **11A**. The posts **11C** each extend straight from the first ring **11A** to the second ring **11B**.
- (13) Each post **11C** has a proximal end that is connected to the second ring **11B** and a distal end that is connected to the first ring **11A**. In this case, the distal end of the post **11C** is connected to the outer surface of the first ring **11A**. The proximal end of the post **11C** is connected to the surface of the second ring **11B** that is opposite to the first ring **11A**. The diameter of the second ring **11B** is greater than the diameter of the first ring **11A** so that the body **11** has a truncated conical form. Thus, the inlet portions **11D**, which are located on the posts **11C**, are positioned outward from the mesh **12** in the radial direction of the first ring **11A**.
- (14) The inlet portions **11D** are, for example, marks indicating where the gates were located when the insert-molded filter **10** was manufactured. In this case, the inlet portions **11D** are gate marks formed by the resin hardened in the gates connected to the molding portion where the body **11** is molded.

- (15) FIG. 2 shows the insert-molded filter **10** as viewed from a view point that is opposed to the opening **11E2** in the first ring **11A**.
- (16) As shown in FIG. 2, the body **11** includes a plurality of inlet portions **11D**. Each inlet portion **11D** is one of the plurality of inlet portions **11D**. Each inlet portion **11D** is located on the corresponding one of the posts **11C**. That is, the posts **11C** each include one inlet portion **11D**. This allows the portion of the tube end of the mesh **12** bent inward in the radial direction of the first ring **11A** to be spread in the circumferential direction of the first ring **11A**.
- (17) In the example shown in FIG. 2, the body **11** includes four posts **11C** and the same number of inlet portions **11D**. The four posts **11C** are arranged at equal intervals in the circumferential direction of the first ring **11A**. Thus, the inlet portions **11D** are arranged at equal intervals in the circumferential direction of the first ring **11A**. In contrast with when the inlet portions **11D** are arranged at irregular intervals in the circumferential direction of the first ring **11A**, uneven distribution of resin having high energy does not occur at the tube end of the mesh **12** in the circumferential direction of the first ring **11A**. Thus, portions in the tube end of the mesh **12** bent inward in the radial direction of the first ring **11A** are not concentrated in the circumferential direction of the first ring **11A**.
- (18) The first ring **11A** includes a wide portion **11AW** that is wider than other parts of the first ring **11A** as viewed from a view point that is opposed to the opening **11E2** of the first ring **11A**. The wide portion **11AW** is located in a circular area defined by an outer edge **11AE** of the first ring **11A**. The bottom wall **12B** of the mesh **12** is embedded in the wide portion **11AW** of the first ring **11A** over an amount corresponding to the wide portion **11AW** so that the mesh **12** further resists separation from the first ring **11A**.
- (19) FIG. 3 shows the insert-molded filter **10** as viewed from a view point that is opposed to one of the posts **11C** of the body **11**.
- (20) As shown in FIG. 3, in the direction in which the first ring **11A** and the second ring **11B** are arranged, the inlet portions **11D** are located on the first ring **11A** at the side that is opposite to the second ring **11B**. Thus, resin forced through the gates having high energy easily reaches the tube end of the mesh **12** and further ensures bending of the tube end of the mesh **12**.
- (21) When the direction in which the first ring **11A** and the second ring **11B** are arranged is referred to as the vertical direction and the first ring **11A** is located upward from the second ring **11B**, the inlet portions **11D** are located upward from the first ring **11A**. In the example shown in FIG. 3, the inlet portions **11D** are located on the distal ends of the arranged posts **11C**. Thus, when the insert-molded filter **10** is manufactured, the gates are located upward and radially outward from the tube end of the mesh **12**. Thus, the tube end of the mesh **12** is positioned in the passage of resin forced through the gates into the molding portion of the mold, and the tube end of the mesh **12** is supplied with resin having high energy. This bends the tube end of the mesh **12** inward in the radial direction of the first ring **11A**.
- (22) FIG. 4 shows an example of the mesh **12** used to manufacture the insert-molded filter **10**.
- (23) As shown in FIG. 4, the mesh **12** has a truncated conical form. The mesh **12** includes the tubular portion **12A**, the bottom wall **12B**, and a connection portion **12C**. The tubular portion **12A** is shaped when two bonding margins positioned at the two ends of a band-shaped portion are bonded to each other. Welded portions **12D** are arranged on the bonding margins in the direction in which the tubular portion **12A** extends.
- (24) The connection portion **12C** connects part of the bottom wall **12B**, more specifically, part of the edge of the bottom wall **12B**, to the tubular portion **12A**. In the tubular portion **12A**, only the part of the tube end that is covered by the bottom wall **12B** is connected to the bottom wall **12B**. Thus, when the insert-molded filter **10** is manufactured, resin is supplied toward the tube end of the tubular portion **12A** so that the tube end of the tubular portion **12A** is bent inward in the radial direction of the tubular portion **12A**.
- (25) The tube end of the tubular portion **12A**, the bottom wall **12B**, and one gap **12E** formed by the

connection portion **12C** are positioned in each of the two sides of the connection portion **12C**. The wide portion **11AW** of the first ring **11A** covers the two gaps **12E**.

(26) A method for manufacturing the insert-molded filter **10** includes preparing the insert-molded mesh **12** described above, and arranging the insert-molded mesh **12** in a mold. The method for manufacturing the insert-molded filter **10** further includes forcing resin from the gates, which are located outward from the tubular portion **12A** in the radial direction of the tubular portion **12A**, into a molding portion of the mold where the body **11** is molded integrally with the mesh **12**.

(27) FIG. 5 schematically illustrates the flow of resin when the insert-molded filter **10** is manufactured. To facilitate illustration and description, FIG. 5 shows the flow of resin during manufacturing in a partial cross-sectional view of the completed insert-molded filter **10**. FIG. 5 also shows a partial cross-sectional view of one of the posts **11C** in the direction in which the posts **11C** is extended and a partial cross-sectional view of the mesh **12**.

(28) As shown in FIG. 5, resin, forced through the gates corresponding to the inlet portions **11D** into the molding portion of the mold, is forced from the outer side of the mesh **12** to the molding portion. This allows the resin forced to the molding portion to flow toward the tube end of the mesh **12**. Further, the resin flows from the portions corresponding to the distal ends of the posts **11C** toward the tube end of the mesh **12**. Thus, in contrast with when the distance is long from the gates to the tube end of the mesh **12**, resin having high energy is supplied to the tube end of the mesh **12**.

(29) When the mesh **12** is arranged in the mold, the tube end that extends straight in cross section in the direction in which the tubular portion **12A** extends as shown by the long-dash double-short-dash line in FIG. 5 is bent inward in the radial direction of the mesh **12** by the resin drawn into the molding portion.

(30) FIG. 6 is a cross-sectional view of the insert-molded filter **10** taken along a plane that includes no posts **11C** and is orthogonal to a plane extending across the opening **11E2** in the first ring **11A**.

(31) As shown in FIG. 6, the tube end of the mesh **12** is bent inward in the radial direction of the first ring **11A** even at positions spaced from the inlet portions **11D** in the circumferential direction of the first ring **11A**. This increases the separation resistance of the mesh **12** from the body **11**.

(32) In the example shown in FIG. 6, among the two tube ends of the mesh **12**, the tube end that is in contact with the second ring **11B** contacts the inner surface of the second ring **11B**. In contrast, the tube end that is in contact with the first ring **11A** contacts the outer surface of the first ring **11A** if it is not bent inward in the radial direction of the first ring **11A**.

(33) When using the insert-molded filter **10**, fluid is supplied to the insert-molded filter **10** from the opening in the second ring **11B**. Thus, fluid flows from the opening in the second ring **11B** toward the openings **11E1** partitioned by the posts **11C**. As a result, force acts on the tube end contacting the second ring **11B** in the direction pressing the tube end toward the inner surface of the second ring **11B** so that the tube end will not separate from the second ring **11B**. In contrast, force acts on the tube end contacting the first ring **11A** in the direction that separates the tube end from the outer surface of the first ring **11A** so that the end tube will easily separate from the first ring **11A**. In this respect, with the insert-molded filter **10**, the tube end of the mesh **12** contacting the first ring **11A** is bent inward in the radial direction of the first ring **11A** and embedded in the first ring **11A**. This increases the separation resistance of the mesh **12** from the first ring **11A**.

(34) As described above, the insert-molded filter according to one embodiment has the following advantages. (1) Resin forced through the gates bends the tube end of the mesh **12** inward in the radial direction of the first ring **11A** so that the tube end of the mesh **12** is at least partially embedded in the first ring **11A**. This ensures the separation resistance of the mesh **12** from the body **11**. (2) The positions of the gates do not impose restrictions on the position of the mesh **12** in the insert-molded filter **10**. This increases the degree of freedom for the position of the mesh **12** in the insert-molded filter **10**. (3) Resin forced through the gates and having high energy easily reaches the tube end of the mesh **12** and further ensures bending of the tube end of the mesh **12**. (4) The inlet portions **11D** are located on the posts **11C** and can be easily positioned outward from the mesh

12 in the radial direction of the first ring **11A**. (5) The portion of the tube end of the mesh **12** bent inward in the radial direction of the first ring **11A** can be spread in the circumferential direction of the first ring **11A**. (6) The bottom wall **12B** of the mesh **12** is embedded in the wide portion **11AW** of the first ring **11A** over an amount corresponding to the wide portion **11AW** so that the mesh **12** further resists separation from the first ring **11A**.

(35) The above-described embodiment may be modified as follows.

(36) Bottom Wall

(37) The mesh **12** does not need to include the bottom wall **12B**. In other words, the mesh **12** does not need to cover the opening **11E2** in the first ring **11A**.

(38) First Ring

(39) The entire first ring **11A** may have a width that allows the gaps **12E** to be covered. This increases the area of the bottom wall **12B** of the mesh **12** that is embedded in the first ring **11A** and increases the separation resistance of the mesh **12** from the body **11**.

(40) Inlet Portions

(41) The quantity of the inlet portions **11D** may be less than the quantity of the posts **11C**. In this case, the inlet portions **11D** are located on only some of the posts **11C**. As described above, each post **11C** may include one inlet portion **11D** to increase the area in which the tube end of the mesh **12** is bent inward in the radial direction of the first ring **11A**.

(42) The body **11** may include only one inlet portion **11D**. In this case, when the inlet portion **11D** is positioned outward from the mesh **12** in the radial direction of the first ring **11A**, advantage (1) is obtained.

(43) The inlet portions **11D** do not need to be located on the posts **11C**. For example, the body **11** may include a portion connected to the first ring **11A** and located outward from the first ring **11A**, and the inlet portions **11D** may be located at such a portion. In this case, the inlet portions **11D** are positioned outward from the mesh **12** in the radial direction of the first ring **11A** and obtain advantage (1).

(44) In the direction in which the first ring **11A** and the second ring **11B** are arranged, the inlet portions **11D** may be located on the first ring **11A** on the side facing the second ring **11B**. In this case, the inlet portions **11D** are positioned outward from the mesh **12** in the radial direction of the first ring **11A** and obtain advantage (1).

Claims

1. An insert-molded filter, comprising: a plastic body including a first ring, a second ring, and posts that are connected to the first ring and the second ring, wherein the body includes openings partitioned by the posts between the first ring and the second ring; and a mesh including a tubular portion, the mesh being molded integrally with the body through insert molding to cover the openings partitioned by the posts, wherein the body includes a plastic inlet portion positioned outward from the mesh in a radial direction of the first ring, and wherein in a direction in which the first ring and the second ring are arranged, the inlet portion is located on the first ring at a side that is opposite to the second ring.

2. The insert-molded filter according to claim 1, wherein the inlet portion is located on the post.

3. The insert-molded filter according to claim 1, wherein the inlet portion is one of a plurality of inlet portions, and each of the plurality of inlet portions is located on a corresponding one of the posts.

4. The insert-molded filter according to claim 1, wherein the mesh further includes a bottom wall that covers an opening in the first ring, the first ring includes a wide portion that is wider than other parts of the first ring as viewed from a view point that is opposed to the opening in the first ring, and the wide portion is located in a circular area defined by an outer edge of the first ring.

5. A method for manufacturing an insert-molded filter, the method comprising: preparing an insert-

molded mesh including a tubular portion; arranging the insert-molded mesh in a mold; and forcing resin through a gate, located outward from the tubular portion in a radial direction of the tubular portion, into a molding portion of the mold where a body is molded integrally with the insert-molded mesh, so that in a direction in which a first ring and a second ring of the body are arranged, an inlet portion of the resin is located on the first ring at a side that is opposite to the second ring.

6. An insert-molded filter, comprising: a plastic body including a first ring, a second ring, and posts that are connected to the first ring and the second ring, wherein the body includes openings partitioned by the posts between the first ring and the second ring; and a mesh including a tubular portion, the mesh being molded integrally with the body through insert molding to cover the openings partitioned by the posts, wherein the body includes a plastic inlet portion positioned outward from the mesh in a radial direction of the first ring, wherein the mesh further includes a mesh bottom wall that covers an opening in the first ring, wherein the first ring includes a wide portion that is wider than other parts of the first ring as viewed from a view point that is opposed to the opening in the first ring, and wherein the wide portion is located in a circular area defined by an outer edge of the first ring.

7. The insert-molded filter according to claim 6, wherein the inlet portion is located on the post.

8. The insert-molded filter according to claim 6, wherein the inlet portion is one of a plurality of inlet portions, and each of the plurality of inlet portions is located on a corresponding one of the posts.
